

Problem B. Another Turtle Problem

Input file: standard input
Output file: standard output

To prevent bthero's tasks from appearing in the contest, the Great Turtle decided to steal the shortlist of problems for the Zhaitykov Olympiad 6202. The jury learned of the plan and set traps in advance; however, the turtle's route is unknown, and trap activations must be programmed before it appears.

The jury office is a rectangular grid of size $n \times m$. Rows are numbered from top to bottom from 1 to n , and columns from left to right from 1 to m .

At time $t = 1$, the turtle appears in cell $(1, 1)$. Then, each time step (in one time unit), it makes exactly one move:

- either one cell down: $(x, y) \rightarrow (x + 1, y)$
- or one cell right: $(x, y) \rightarrow (x, y + 1)$

until it reaches cell (n, m) . Thus, there are exactly $T = n + m - 1$ time moments $t = 1, 2, \dots, T$, and at time $t = T$ the turtle is in cell (n, m) .

The jury has placed k traps. Each trap is a square area described by four numbers:

- x_i, y_i — coordinates of the **top-left** cell of the trap area (with the smallest row and column indices),
- s_i — the side length of the square area,
- $cost_i$ — the cost of **one activation** of the trap.

Trap i covers all cells (a, b) such that:

$$x_i \leq a \leq x_i + s_i - 1, \quad y_i \leq b \leq y_i + s_i - 1$$

It is guaranteed that the trap area lies entirely inside the grid.

Due to the "net-based" mechanism, a trap is successful only if it is activated precisely when the turtle is standing on a cell covered by the trap. The jury may activate any trap i at any time moment; each activation at each time moment costs $cost_i$. Multiple traps may be activated simultaneously.

Since the turtle's route is unknown, you must beforehand choose at which moments to activate which traps so that for any possible route, the turtle will be caught.

You must find the minimum total cost of such an activation schedule, or determine that it is impossible to guarantee catching the turtle.

Input

In the first line three integers n, m, k are given ($2 \leq n, m \leq 10^5$, $1 \leq k \leq 10^5$) — the grid dimensions and the number of traps.

In the next k lines, the trap descriptions are given. Line i contains four integers $x_i, y_i, s_i, cost_i$ ($1 \leq x_i \leq n$, $1 \leq y_i \leq m$, $0 \leq cost_i \leq 10^9$) — the coordinates of the top-left cell, the side length of the square, and the cost of one activation.

Output

Output one integer — the minimum total cost required to guarantee catching the turtle for any valid route.

If it is impossible, output -1 .

Scoring

Subtask	Additional Constraints	Points
0	Examples	0
1	$n = 2$	11
2	$k \leq 10, n, m \leq 100$	9
3	$cost_i = 0, n, m \leq 1000$	10
4	$cost_i = 0, k \leq 1000$	16
5	$k \leq 1000$	24
6	$n, m \leq 1000$	14
7	Without additional constraints	16

Examples

standard input	standard output
2 3 2 1 2 2 5 2 1 1 3	5
8 8 4 1 2 2 3 4 1 4 10 2 5 3 4 1 7 1 10	17

Note

In the first sample, we can activate the first trap at time $t = 3$. One can notice that in this case, we will catch the Turtle for sure.